MATHS AA HL · Trigonometric symmetry and identities

Worked mark scheme · 25 minutes · 18 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

1. Reflections and a hidden root [10]

(a) **[3]**

Use unit-circle symmetry to simplify $F(x)$.

R1 · Reflection across the vertical axis preserves the sine coordinate.

R1 · It reverses the cosine coordinate.

A1 · Hence $F(x) = 2(\sin x + \cos x)$.

(b) **[3]**

Solve $F(x) = 2$ exactly, retaining both endpoints where appropriate.

M1 · Write $\sin x + \cos x = \sqrt{2}\sin(x + \pi/4)$.

M1 · Solve $\sin(x + \pi/4) = 1/\sqrt{2}$.

A1 · The solutions are 0, $\pi/2$ and 2π.

(c) **[4]**

Find all real k such that $F(x) = k$ has exactly three distinct solutions on the stated closed interval. Justify completeness.

M1 · Its range is $[-2\sqrt{2}, 2\sqrt{2}]$.

R1 · On a half-open period every interior ordinate is taken twice, with just one root at an extremum.

R1 · Closing the interval adds a second endpoint only at the ordinate $F(0) = 2$.

A1 · Hence $k = 2$ is the unique value.

2. Which transformations preserve a trigonometric equation? [8]

(a) **[4]**

Show that $g(\pi - x) = g(x)$ and $g(x + \pi/2) = -g(x)$.

M1 · Under the first reflection the cosine changes sign but both squares are unchanged.

A1 · Hence $g(\pi - x) = g(x)$.

M1 · Under the shift the sine and cosine squares exchange roles.

A1 · This gives $-g(x)$.

(b) **[4]**

Solve $g(x) = 1/2$ on $0 \le x < 2\pi$. A student argues that shifting each solution by $\pi/2$ supplies four more. Evaluate the argument.

M1 · Use $g(x) = -\cos(2x)$.

M1 · Solve $\cos(2x) = -1/2$ over two full periods.

A1 · $x = \pi/3, 2\pi/3, 4\pi/3, 5\pi/3$.

R1 · The shift maps these to solutions of $g(x) = -1/2$, so it adds none.

End of mark scheme · 18 marks